



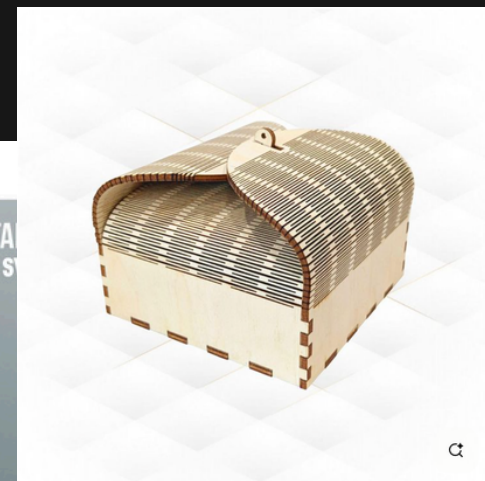
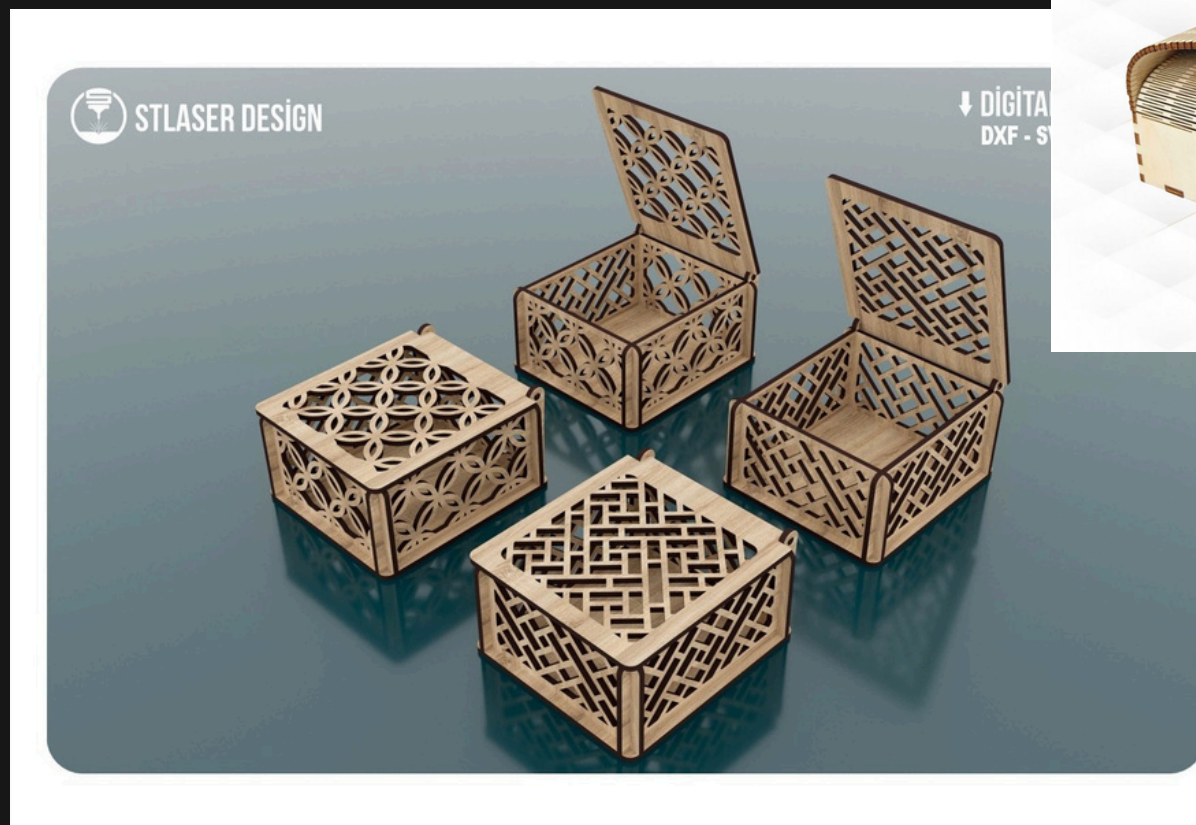
ARDUBOY



BOX

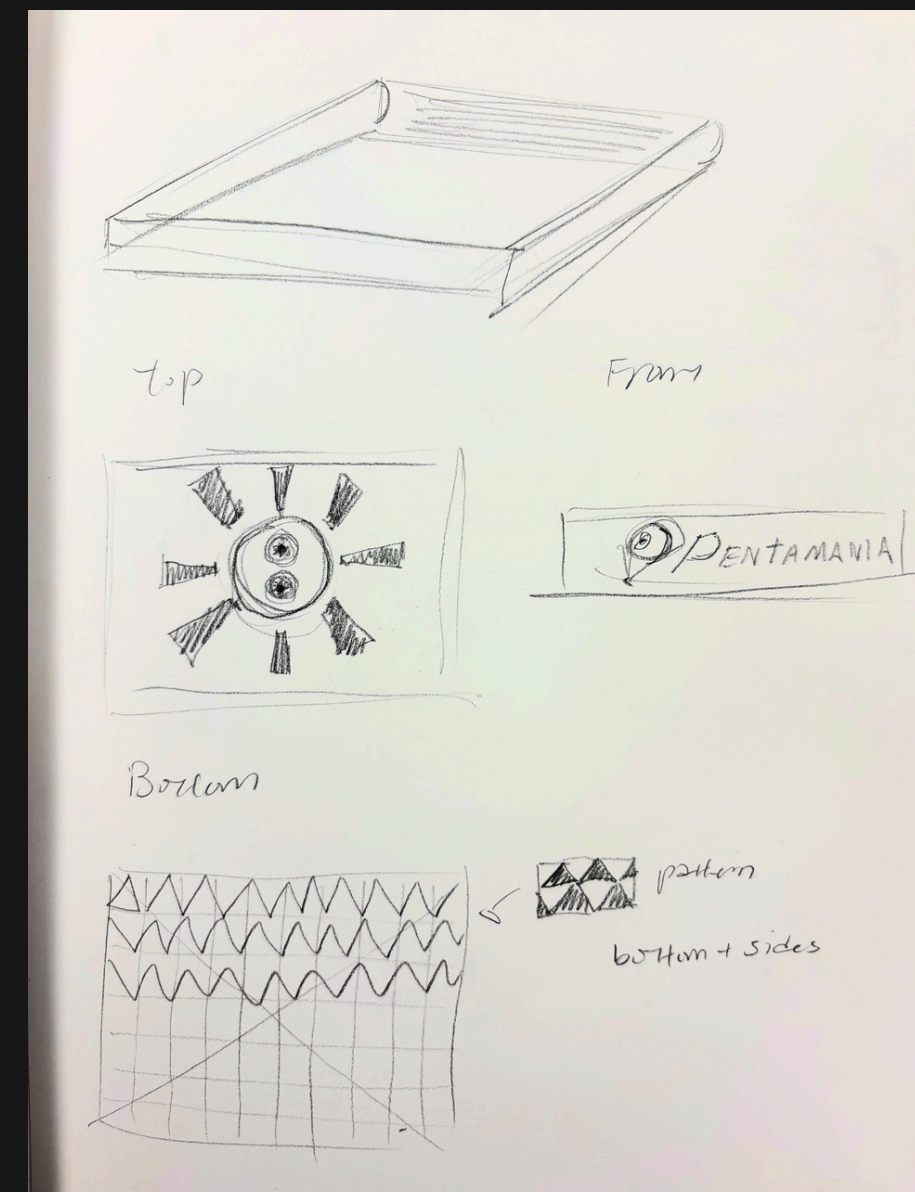
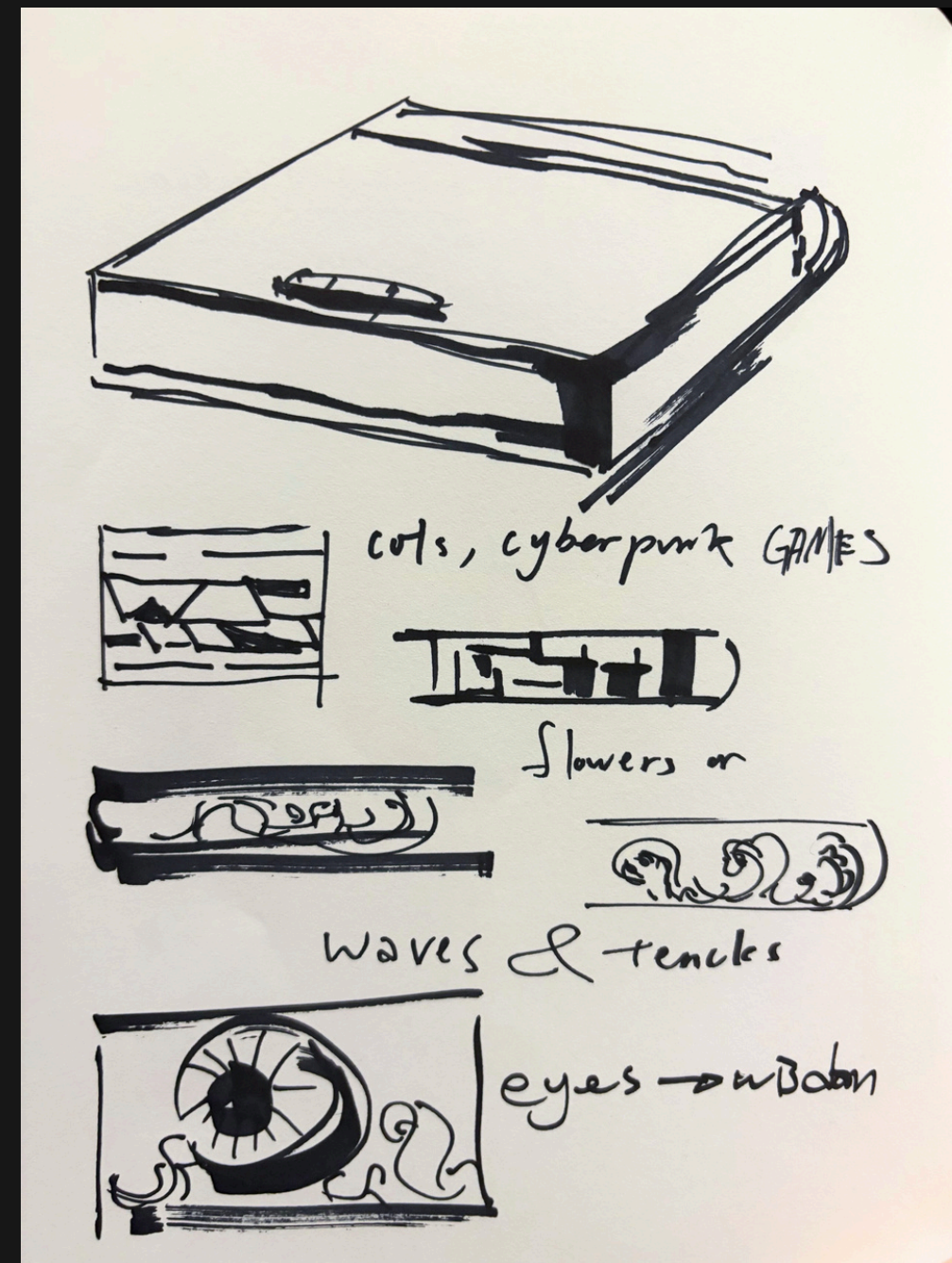
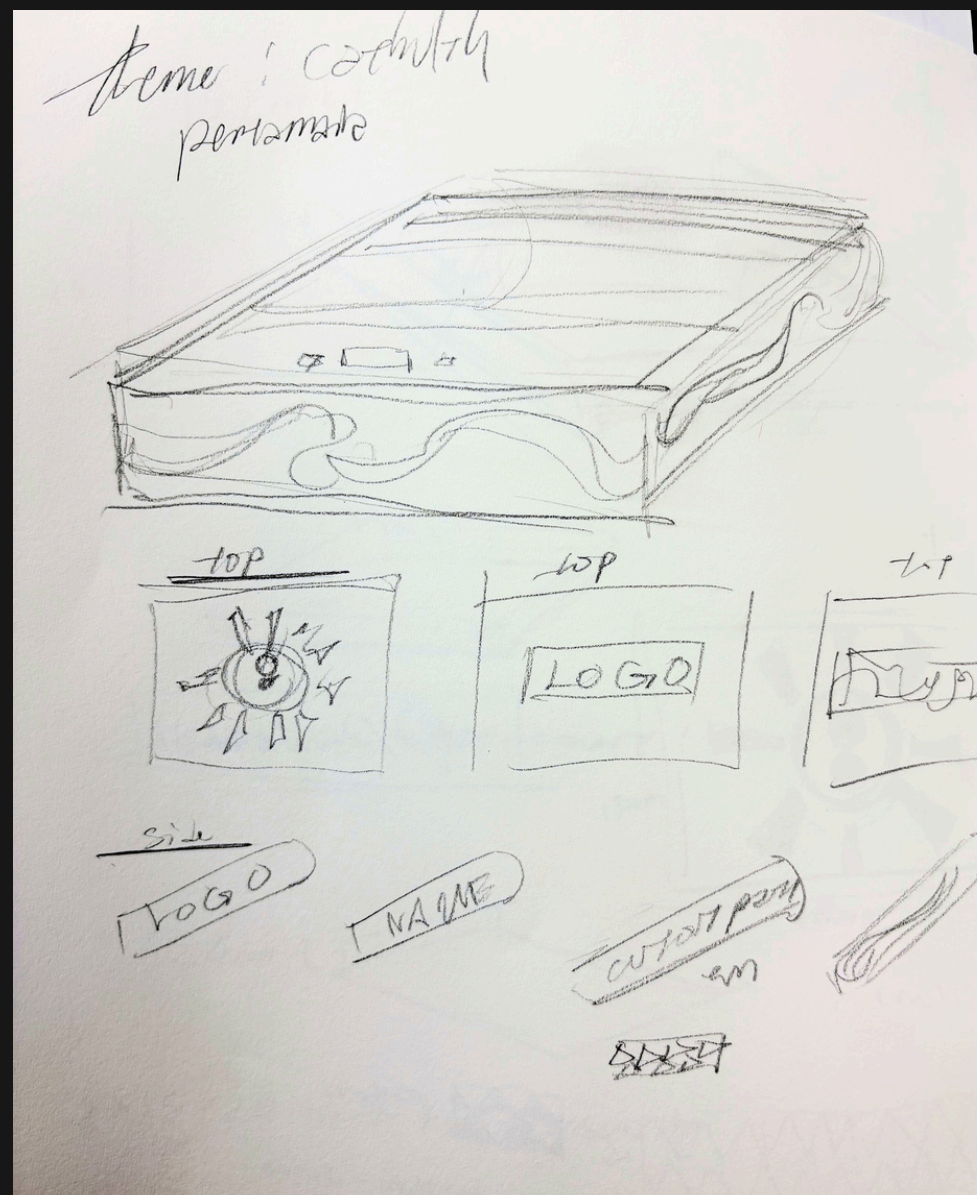
MOODBOARD & CONCEPT DEV

This collection process was aimed at searching for both time efficient and simple designs for deadline constraints. I know I want lots of engraving and incorporating the living hinge so my focus was on similar designs.



ROUGH CONCEPT SKETCHES

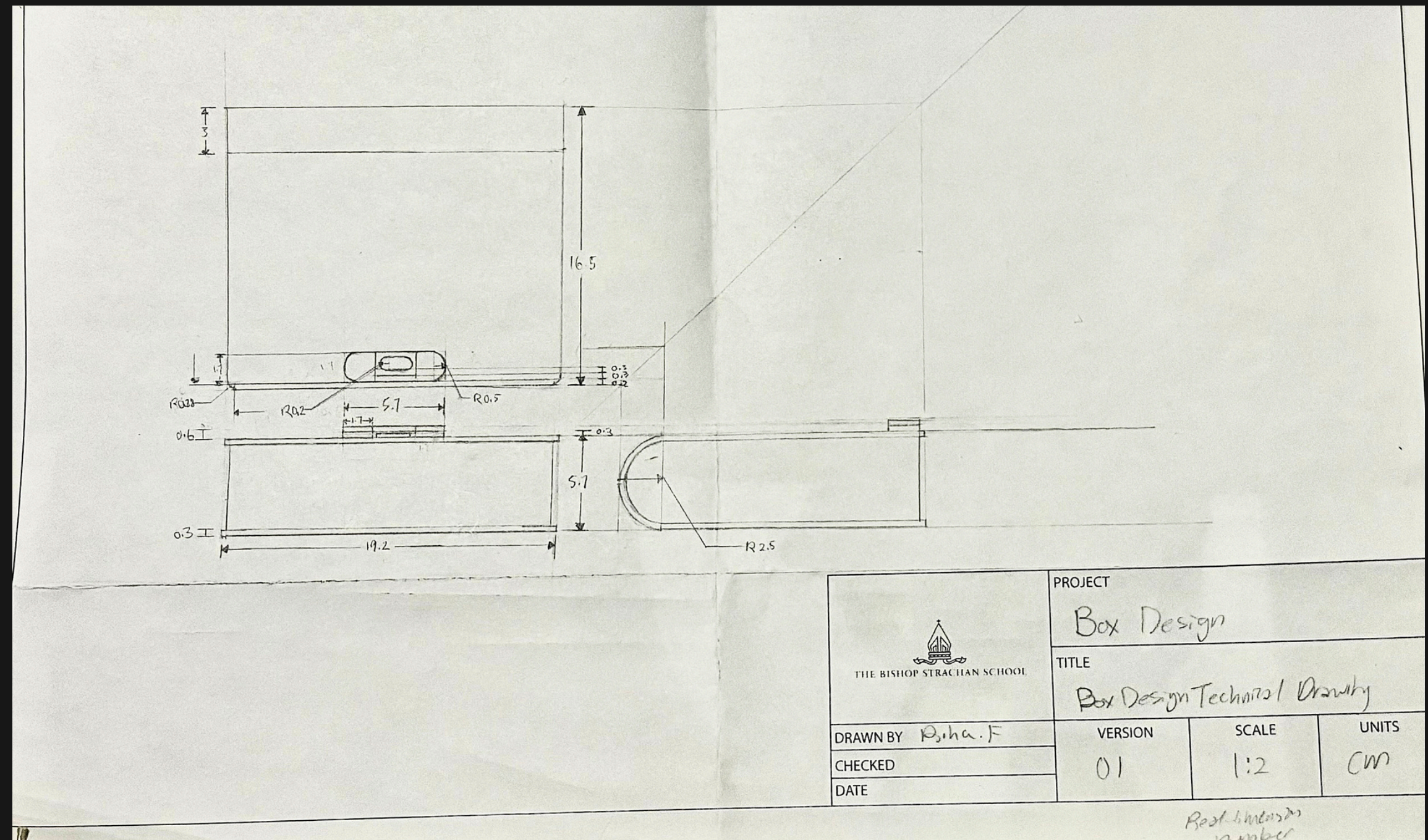
These are some rough process of my initial ideas. Since clean & accurate drawings aren't my advantages, I moved neater revisions in to the illustrator design process. That being said, some patterns and placement stayed to my final design (p3)



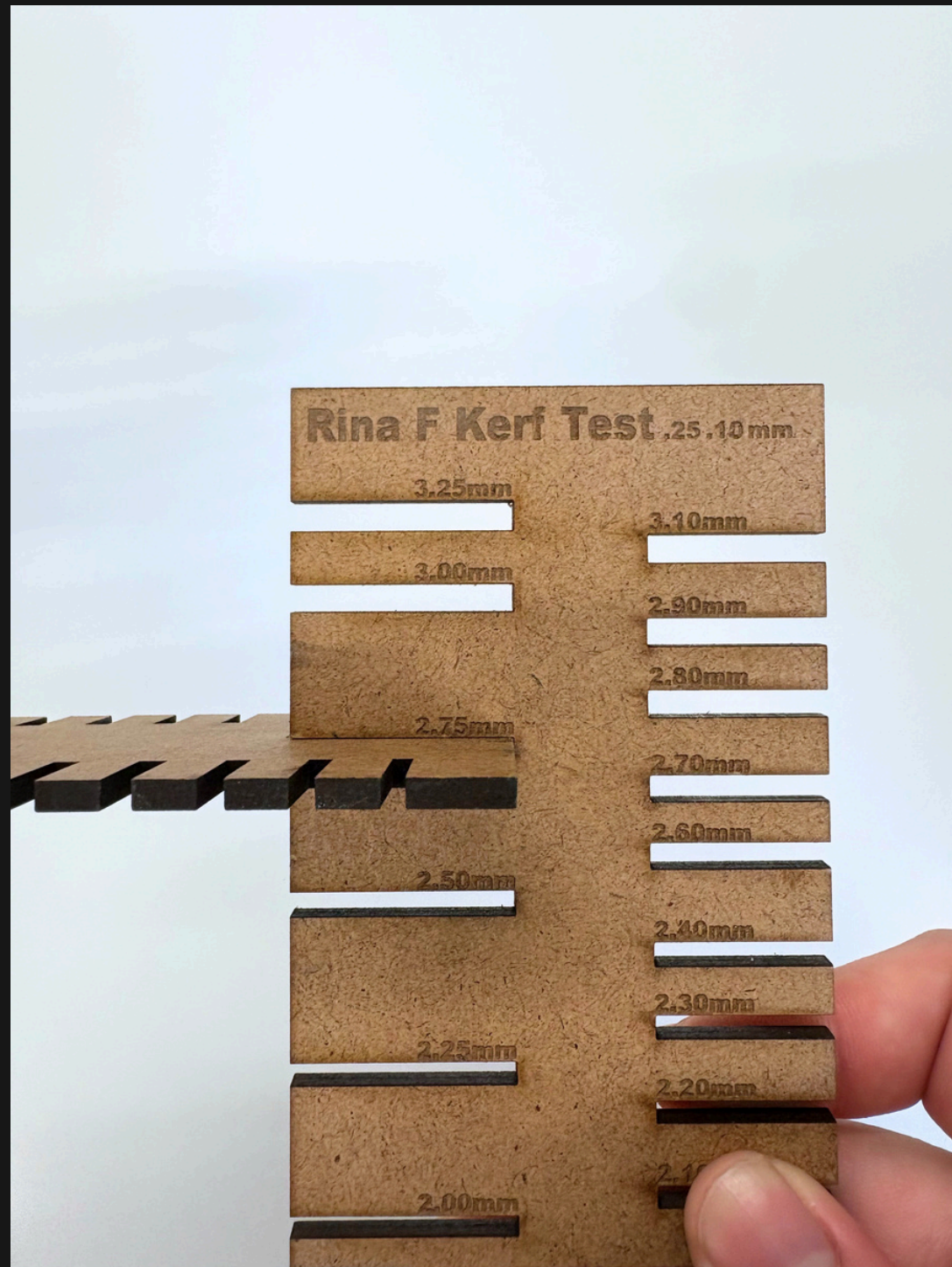
ORTHOGRAPHIC DRAWING

My orthographic drawing ignored tabs, living hinges and hidden lines.

(PS: Because it was folded the lines looks a bit tilted.



KERF TEST.



/2

Fit & Laser Behaviour

The kerf test tool helped me evaluate how much extra material the laser removed and how that affects joint fit. At tighter tolerances, joints required noticeable force to assemble, which increased the risk of cracking. This test showed that laser power and material thickness directly affect fit, and that kerf compensation is necessary to achieve accurate press-fit joints rather than relying on nominal dimensions.

Rina F Kerf Test .25 .10 mm

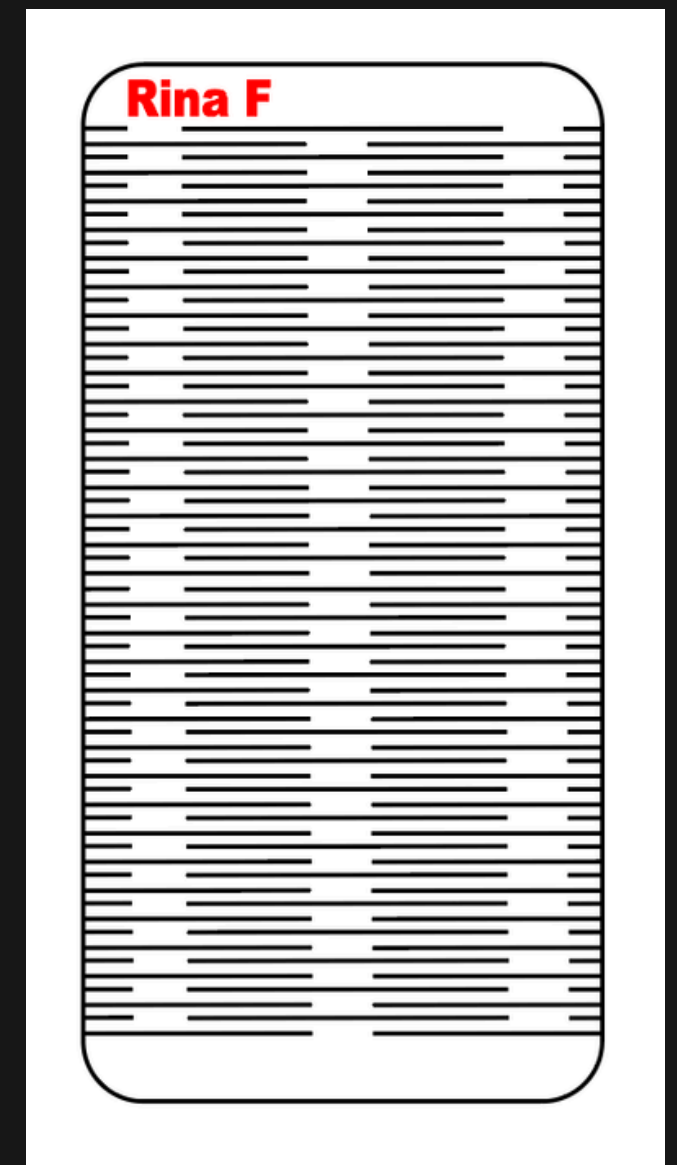
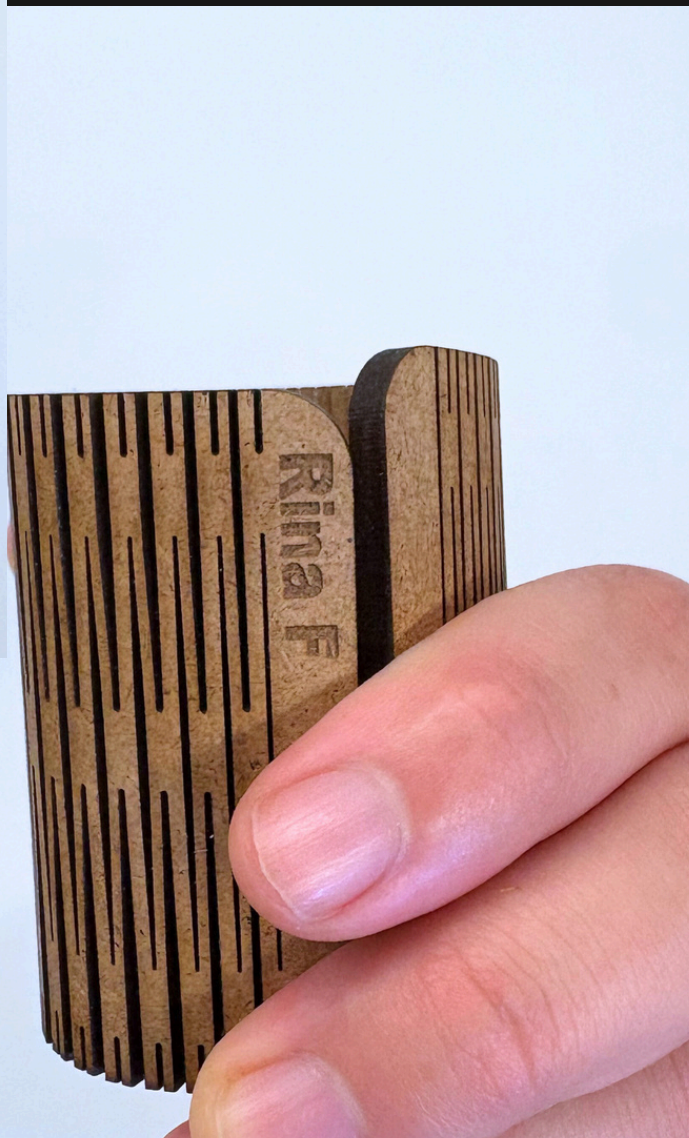
3.25mm	3.10mm
3.00mm	2.90mm
	2.80mm
2.75mm	2.70mm
	2.60mm
2.50mm	2.40mm
	2.30mm
2.25mm	2.20mm
	2.10mm
2.00mm	

HINGE TEST.

/2

Flexibility, Strength, and Geometry

- Flexibility: The thinner the row spacing the more smooth bending gets. I think this factor is the most important in deciding flexibility. Personally I prefer a more flexible hinge so I choose a 1.5mm spacing.
- Strength: One interesting discovery I made about strength is the difference between the top and bottom side of the hinge. Turns out the bottom side is actually more sturdier because less material is burnt away.
- Best Geometry: The line pattern naturally works well for this project because we only require it to bend one way compared to other complex geometry patterns that can bend in other directions.



/2



/3



/2 Front engraving with logo (different depth)

/3 Side engraving showing a continuation of pattern

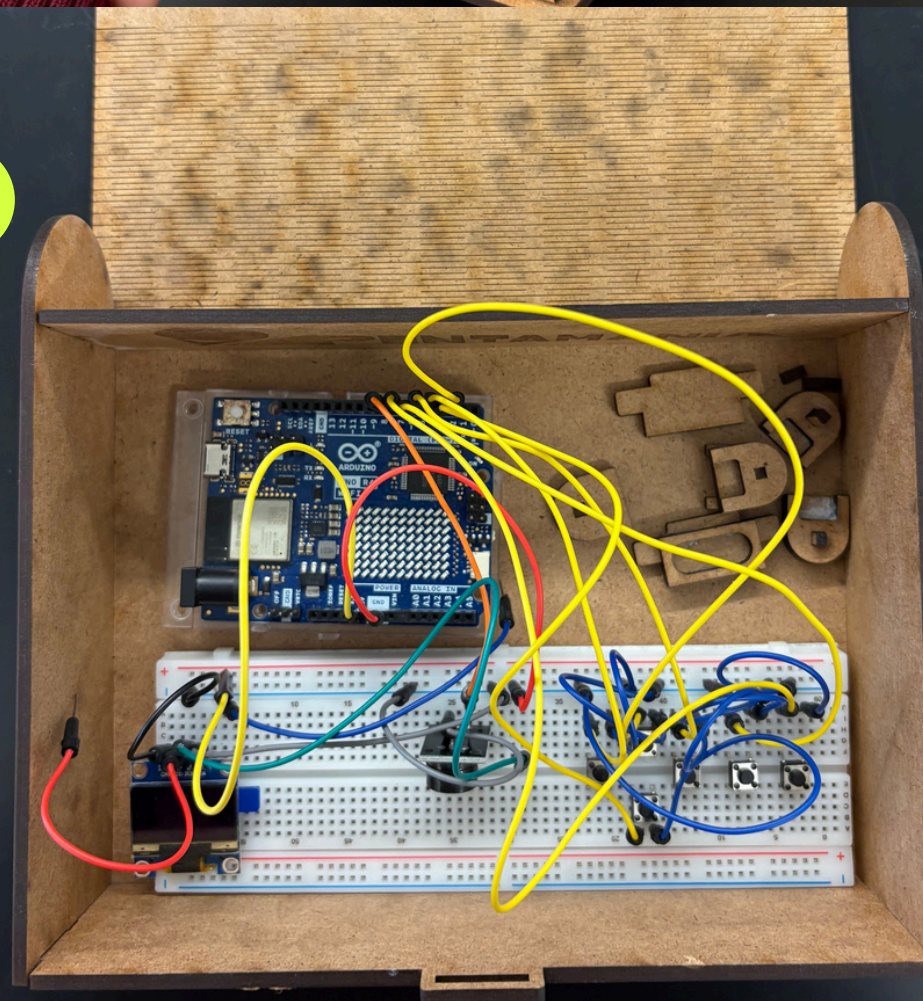
/4 Inside engraving with logo

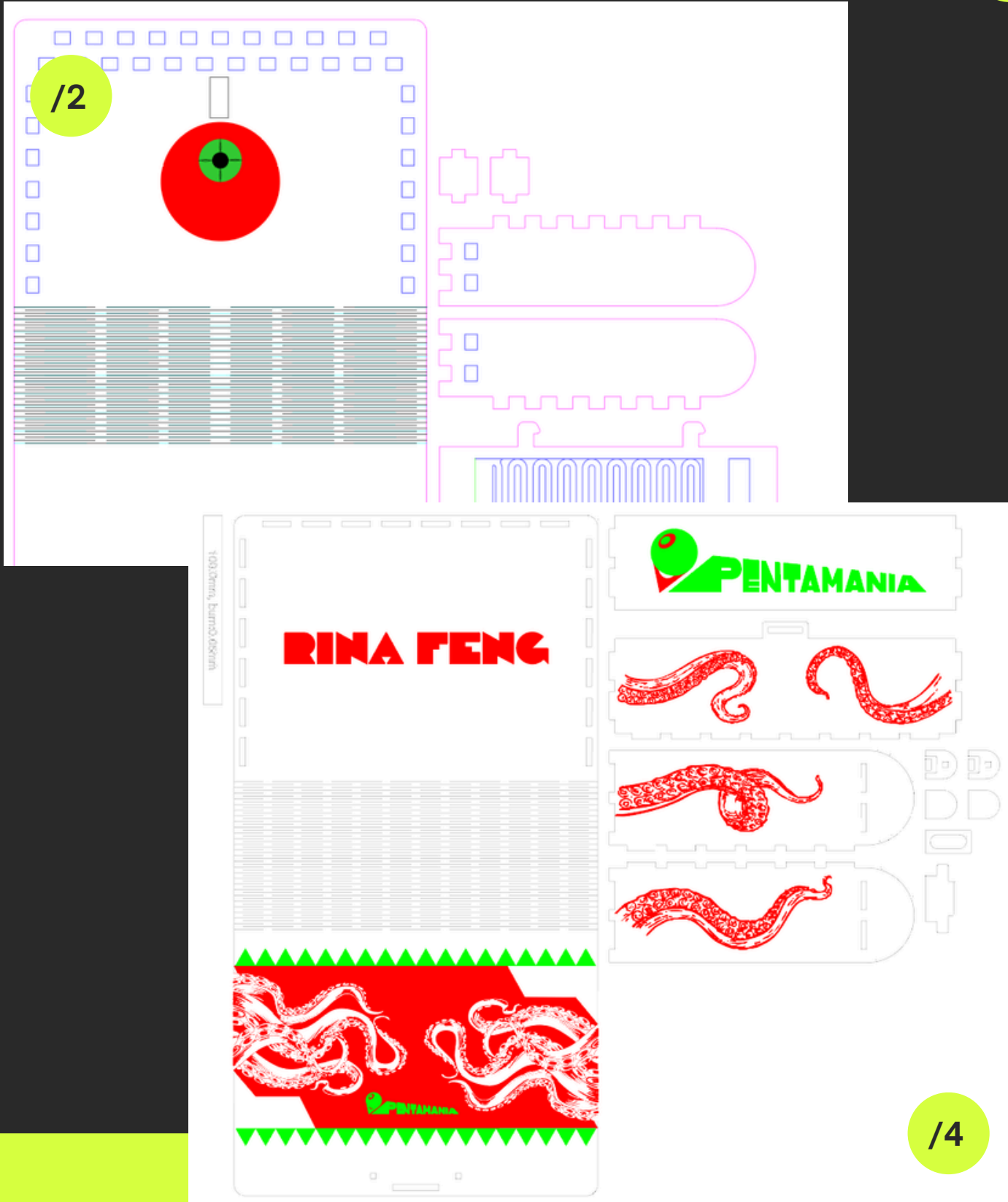
/5 Snug fit of the Arduboy with wiggle room

/4

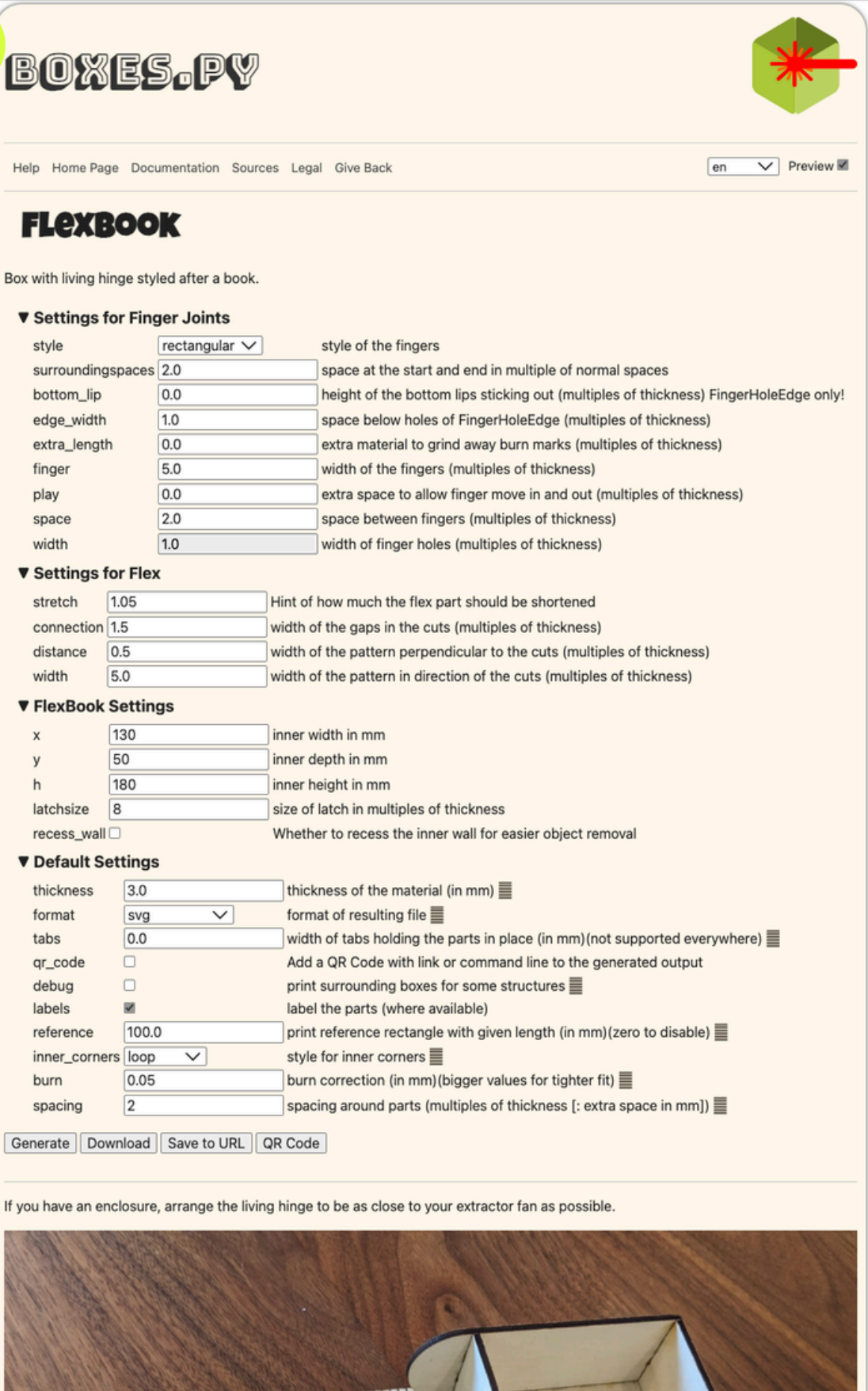


/5





/3



/2

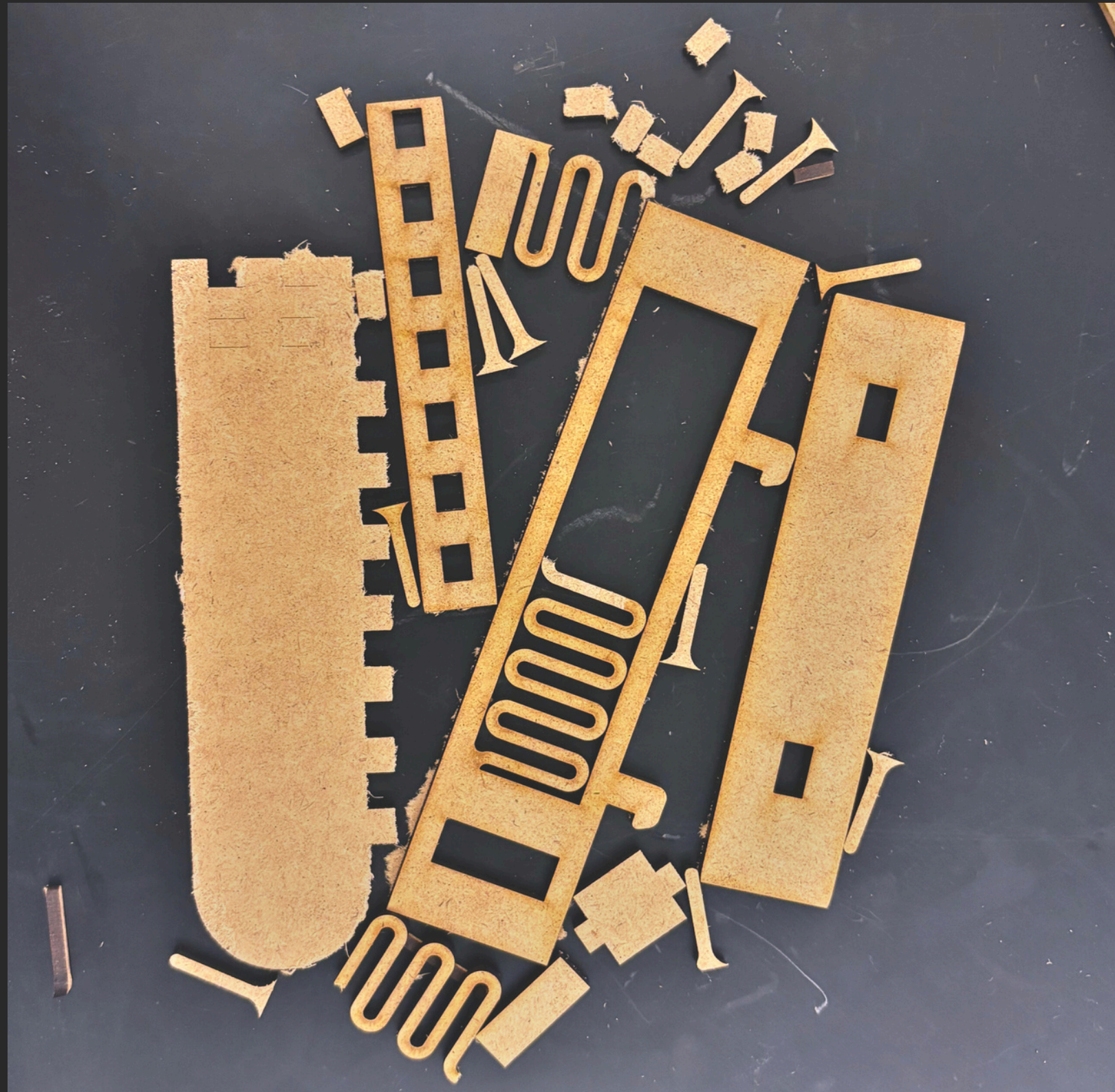
One of many failed designs. Scale + kerf issue, wrong elasticity, wrong dimensions. .

/3

Final dimensions from Boxes.py.

/4

Final design.

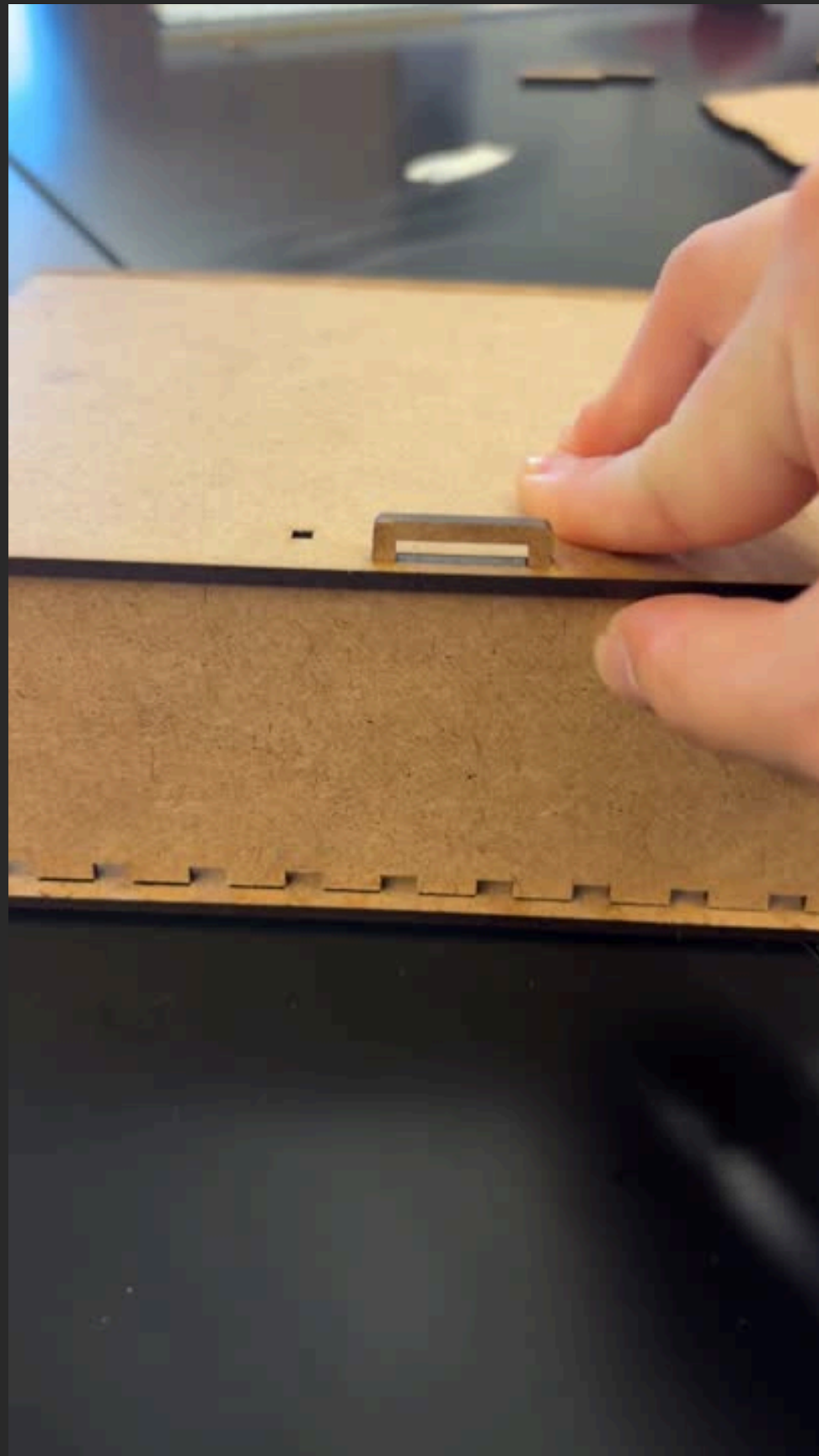


DESIGN CRITERIA EVALUATION:

The hardest design criteria to meet was finding the correct **kerf value**. This was extremely hard in two aspects. **First**, different sources suggests different values, and it takes many tries to find the perfect value tight enough to assemble without glue, but not too tight that it doesn't fit. **Second**, the level of precision required made it hard to identify what went wrong when something failed. For example, even with physical kerf tests, I am still betting on a 0.05mm difference that could be a total failure or a perfect success. Especially in some cases, my experiments, Illustrator file, and measurements all suggested the fit should work, yet it didn't, which makes it extremely difficult to determine what exactly went wrong. This really made me balance between **precision and time**, which means testing out the perfect kerf value but suffering with a unoriginal design.

CLIENT JUSTIFICATION:

To be honest, even though I really like the look of engravings, not including them was the better choice when balancing usability, laser-cutting time, and assembly time. First, engravings like mine that are especially large and multi dramatically increase laser-cutting time. In my case, they increased the cutting time by about **450%**, from around **17 minutes to over 1 hour and 20 minutes**. This made the production process much less efficient. Second, engravings also affect assembly because they create more dust and residue, which requires extra cleaning. Removing this feature allowed me to focus on function and structural fit rather than spending time on details that did not improve the box's main purpose.



0.05?

0.2?

0.1?

0.25?

0.15?

0.178?

KERF COMPENSATION ANALYSIS:

Kerf was one of the biggest challenges I faced in this project, mainly because I kept getting different values from different sources. My kerf testing tool suggested a value of 0.25 mm, but when I used 0.2 mm for my first test to be safe, it failed completely (see video). Then, my friend tried values like 0.178 mm and 0.1 mm using the same template from the same website, and those still didn't press-fit properly. In the end, I entered **0.05** mm, expecting it to fail (so I can glue it together), but surprisingly, it worked. This experience showed me how **unpredictable** laser cutting can be and how even very small numerical changes can drastically affect the final fit. It also made me realize that real-world experiments does not always match theory:(

MARKETABILITY & VALUE PROPOSITION:

This box provides value that mass-produced storage boxes cannot because it was designed specifically for Arduboy components, which ensures it fits **securely**, preventing movement damage. It also has a **lockable** feature, which increases protection and makes the box safer for transport and long-term storage. Lastly, the **engravings** it adds aesthetic value and makes the box it's self a work of art, turning it into more than just a container. The balance between security, usability, and aesthetics makes this box both practical and valuable, which increases its long-term value compared to a standard, factory-made container.



PORTFOLIO MANAGEMENT (WHY SHOW YOUR PROCESS?):

Including the lab exploration in the portfolio is quite important because it shows the process of **revision thinking** and not just a final design. The kerf and hinge tests demonstrate clear focused **experimentation** and **problem-solving**. This kind of documentation proves the student understands the process of design and can apply design thinking in the future, capable of making actual **innovations**. For universities or companies, the ability innovate and revise is more important than just showing the technical capability.

Rina Feng



Presentation



THANK YOU